

Architecture Decision Record Register

Living Document — Updated continuously throughout all 10 design steps

Document ID	ADR-REG-001
Version	2.0 — Step 6 complete. 7 ADRs resolved. 3 APPROVED. 4 IN PROGRESS pending stakeholder sign-off.
Date	07 June 2026
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ADRs in register	7 — 3 APPROVED. 4 IN PROGRESS.

1. Purpose and Usage

This register tracks every Architecture Decision Record (ADR) raised during the RetailSphere Data Platform Programme. An ADR documents a significant architectural decision — one that is difficult to reverse, has material impact on multiple domains, or requires formal stakeholder sign-off before implementation proceeds.

This register is a living document. New ADRs are added as significant decisions are identified during any step of the 10-step design process. The individual ADR documents — containing full context, options considered, decision made, and consequences — are written when the relevant step is reached. The register provides visibility of the full decision backlog at all times.

How to read the status column:

- PENDING — Decision identified but not yet made. ADR document not yet written.
- IN PROGRESS — ADR document being drafted. Options under active evaluation.
- APPROVED — Decision made, documented, and signed off by approver. Implementation may proceed.
- SUPERSEDED — Decision was made but has been replaced by a subsequent ADR.

2. ADR Summary Register

ADR ID	Title	Triggered By	Status	Owner	Approver
ADR-001	Master Product Identifier Strategy	BDR-001 v1.0	APPROVED	T. Mbalo (DSA)	N. Mahlangu (CDO)

ADR ID	Title	Triggered By	Status	Owner	Approver
ADR-002	Unified Customer Identity Resolution Strategy	BDR-001 v1.0	IN PROGRESS	T. Mbalo (DSA)	N. Mahlangu (CDO) + Legal (POPIA)
ADR-003	Supplier Contract Data Extraction Strategy	Step 2	IN PROGRESS	T. Mbalo (DSA)	N. Mahlangu (CDO) + Siphon Dlamini
ADR-004	Medallion Architecture Layer Design	Design philosophy agreed	APPROVED	T. Mbalo (DSA)	N. Mahlangu (CDO)
ADR-005	Orchestration and Pipeline Strategy	Design philosophy agreed	APPROVED	T. Mbalo (DSA)	N. Mahlangu (CDO)
ADR-006	Marketing Attribution Logic and Methodology	Step 2	IN PROGRESS	T. Mbalo (DSA)	N. Mahlangu (CDO) + Rethabile Sithole (CCO)
ADR-007	Contact Governance and Opt-out Propagation	Step 2 / BDR-001 v1.2.	IN PROGRESS	T. Mbalo (DSA)	N. Mahlangu (CDO) + Priya Naidoo (Head of CRM & Loyalty)

3. ADR Detail Summaries

Step 6 ADR Resolution is complete. All 7 ADRs have been formally documented with context, options, decision direction, and consequences. 3 ADRs are APPROVED and ready for implementation. 4 ADRs are IN PROGRESS pending stakeholder and legal sign-off before implementation can proceed. Step 9 Physical Database Design depends on ADR-001 and ADR-002 being fully APPROVED.

ADR-001 — Master Product Identifier Strategy

Status	APPROVED – 06 June 2026
Design Step	Step 6 — Risk Analysis and ADR Resolution
Triggered By	BDR step 1 and BPA step 2, where three different systems have three different product codes for the same product.
Owner	T. Mbalo (DSA) — Approver: N. Mahlangu (CDO)
Due	RESOLVED — Step 6
Context	SAP ECC stock code ≠ Oracle MICROS POS product code ≠ Shopify SKU. No master product identifier exists across RetailSphere's three primary transactional systems. Cross-domain product reporting is impossible without resolution. The platform will solve this by creating a surrogate key in the Silver layer Product Dimension by matching the barcode (UPC/EAN) from all systems.
Decision Direction	DECIDED - Product Dimension. Following the fallback matching hierarchy – (1) Barcode match - most reliable. (2) Product name + category match - fuzzy matching. (3) Manual resolution by Siphon's team - human validation queue (same approach as PDF extraction)
Consequences if Delayed	N/A - Decision made.

Dependencies	All downstream steps — Steps 7-18 depend on this decision being locked.
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ADR-002 — Unified Customer Identity Resolution Strategy

Status	IN PROGRESS
Design Step	Step 6 — Risk Analysis and ADR Resolution
Triggered By	BDR-001 v1.0 — Step 1. Scope expanded BDR-001 v1.2.
Owner	T. Mbalo (DSA) — Approver: N. Mahlangu (CDO) + Legal (POPIA)
Due	Pending POPIA Legal review.
Context	Salesforce Loyalty ID ≠ Shopify Customer ID ≠ POS receipt ≠ Marketing Cloud contact ID ≠ Meta pixel ID ≠ GA4 client ID. 66% of in-store transactions are anonymous. 2.1M loyalty members cannot be reliably linked to online or in-store transactions without identity resolution logic.
Decision Direction	DECIDED - Customer Dimension in Silver Layer. Following the fallback matching hierarchy - (1) Deterministic - exact match: ID Number - most reliable, capture it for loyalty. (2) High Confidence: First name + Last name + Phone number. People rarely change names, and cell numbers are relatively stable. (3) Medium confidence (fuzzy): Email address + First name. Email can change, but a first name narrows the match significantly. (4) Low confidence: Phone number + First name only. Weaker - requires human validation before linking identities. (5) Anonymous: 66% of in-store transactions. No identity signal available. Segment as Anonymous in-store, cannot be linked - documented in FR-SAL-003 customer segments.
Consequences if Delayed	Until resolved, no Customer Lifetime Value calculation, no true retention rate, no closed-loop attribution. Affects Sales, Customer Loyalty, and Marketing domains. Customer dimension SCD Type 2 design in Step 7 depends on this decision.
Dependencies	POPIA legal review. Data profiling of email match rate between Shopify and Salesforce TBC from Step 5 investigation. Must be completed before Silver Layer identity resolution is built)—stakeholder sign-off from Priya Naidoo (Head of CRM and Loyalty) and Rethabile Sithole (CCO)..

ADR-003 — Supplier Contract Data Extraction Strategy

Status	IN PROGRESS
Design Step	Step 6 — Risk Analysis and ADR Resolution
Triggered By	Step 2 — Sipho Dlamini interview BPA-001
Owner	T. Mbalo (DSA) — Approver: N. Mahlangu (CDO) + Sipho Dlamini (Head of Procurement)
Due	Pending Sipho Dlamini and N. Mahlangu formal sign-off.
Context	RetailSphere has 340 active supplier contracts stored as PDF documents on a shared drive. These contracts contain contracted lead times, payment terms, and price schedules — data required to calculate price variance against contracted rates (one of Sipho's five Monday morning metrics). pypdf can extract text, but cannot produce structured fields without additional processing.
Decision Direction	DECIDED - LLM-assisted hybrid approach. (1) Text extraction: pypdf extracts raw text from each PDF document and handles standard PDF text layers. (2) LLM Structuring: LLM (Claude or similar) reads the extracted text and identifies key fields: Contracted rate per product, Lead time in days, Payment terms (30/60/90), Supplier name and code, and Contract start and end date. Outputs structured JSON or CSV ready for Bronze layer loading. (3) Human Validation Queue: Low confidence extractions flagged for Sipho's team to review and confirm before loading to Bronze. Prevents incorrect contracted rates from entering the price variance calculation.
Consequences if Delayed	Until resolved, the price variance against contracted rates metric cannot be calculated. This is one of Sipho's five required KPIs. Affects the Procurement domain Gold layer completeness

Dependencies	A sample of 10-15 supplier contracts is required for extraction testing. Procurement team availability for validation queue management.
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ADR-004 — Medallion Architecture Layer Design

Status	APPROVED — 01 June 2026
Design Step	Step 6 — Risk Analysis and ADR Resolution
Triggered By	10 multi-source systems requiring unified architecture across all domains.
Owner	T. Mbalo (DSA) — Approver: N. Mahlangu (CDO)
Due	RESOLVED — Step 6
Context	We have about 10 multi-source systems (different types of data structures) from which we are extracting data. Audit trail requirement, allowing us to reprocess from the Bronze layer, as there will be no need to re-extract from the sources, and only the changed data is being processed, which will save us time in case of failure and the need to reprocess the pipeline. Each layer processes only what has changed since the last run. Bronze captures raw changes. Silver transforms only new records. The Gold layer refreshes only affected aggregates.
Decision Direction	DECIDED — Medallion Architecture chosen. Bronze → Silver → Gold. Reasons: Audit trail (Johan), data governance and POPIA (Priya), 10 multi-source support, incremental load requirements.
Consequences if Delayed	N/A — Decision made.
Dependencies	All downstream steps — Steps 7-18 depend on this decision being locked.

ADR-005 — Orchestration and Pipeline Strategy

Status	APPROVED - 01 June 2026
Design Step	Step 6 - Risk Analysis and ADR Resolution.
Triggered By	Need to define a single orchestration tool before the pipeline build begins in Step 13. Multiple tools were considered during the source system investigation.
Owner	T. Mbalo (DSA) — Approver: N. Mahlangu (CDO)
Due	RESOLVED — Step 6
Context	Azure Data Factory is already used for our data ingestion layer connecting to all 10 source systems to the Bronze Layer. Easy to learn with the drag and drop capabilities, minimal coding required. Easily integrated with all other Azure services.
Decision Direction	DECIDED - Azure Data Factory chosen.
Consequences if Delayed	N/A - Decision made.
Dependencies	All pipeline builds steps — Steps 13-16 depend on the orchestration tool being confirmed. ADF configuration and Azure subscription required before the build starts.

ADR-006 — Marketing Attribution Logic and Methodology

Status	IN PROGRESS
Design Step	Step 6 — Risk Analysis and ADR Resolution

Triggered By	Step 2 — Rethabile Sithole (CCO) interview. BDR-001 v1.2.
Owner	T. Mbalo (DSA) — Approver: N. Mahlangu (CDO) + Rethabile Sithole (CCO)
Due	Pending Rethabile's approval.
Context	RetailSphere spends R8.3M annually across Salesforce Marketing Cloud, Meta Business Manager, and Google Analytics 4. Each platform applies its own attribution model resulting in double and triple counting. Marketing attribution data does not exist in any source system — the data platform will create it for the first time. Attribution window definition and methodology must be agreed before build as it becomes the system of record.
Decision Direction	PENDING — Key decisions required: (1) Attribution model type: last-click, first-click, linear, time-decay, or data-driven. (2) Attribution window: how many days between campaign touchpoint and transaction counts as a conversion. (3) Cross-channel weighting: how to handle a customer who touches multiple channels before converting. (4) Treatment of anonymous-to-known conversion events.
Consequences if Delayed	This is the highest-risk ADR in the register. Once Rethabile Sithole begins making R8.3M annual spend decisions based on platform attribution numbers, changing the methodology becomes politically very difficult. Must be agreed, documented, and signed off before implementation. Rethabile cannot validate numbers against any existing source of truth.
Dependencies	ADR-002 (Customer Identity Resolution) — attribution requires linked customer identity across channels. Rethabile Sithole formal sign-off. Legal review of data use across Meta and GA4 (POPIA implications for pixel-based tracking).

ADR-007 — Contact Governance and Opt-out Propagation

Status	IN PROGRESS
Design Step	Step 6 - Risk Analysis and ADR Resolution.
Triggered By	Step 2 — Priya Naidoo (Head of CRM & Loyalty) interview. BDR-001 v1.2.
Owner	T. Mbalo (DSA) — Approver: N. Mahlangu (CDO) + Priya Naidoo (Head of CRM & Loyalty)
Due	Pending Priya Naidoo's approval
Context	This must be formally agreed and signed before any governance logic is built. Covers tiered ingestion design, suppression logic, breach alerting, and cross-system propagation rules. At-risk member flag - three components required: Last purchase date - from Sales fact table, Last category purchased - derived from Sales fact table joined to Product dimension. Last engagement date - from Contact History fact table.
Decision Direction	PENDING - (1) How does opt-out propagation work technically - what triggers the cascade across all systems. (2) How is the 4-contact cap enforced where does the check happen. (3) Who owns the Contact History fact table governance.
Consequences if Delayed	Members who have legally withdrawn marketing consent continue to receive communications through channels that have not received the opt-out signal. This is a direct POPIA compliance risk.
Dependencies	ADR-002 (Customer Identity Resolution), The following requires ADR-007 to be resolved: NFR-LEG-001 — POPIA Compliance, FR-CUS-001 — Communications per member, and FR-CUS-002 — Opt-out propagation.

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